

Appendix B.6

Appendix B.6.1 Model Specification and Sample Summary

Item	Detail
Unit of analysis	One chatbot response paired with its originating user question.
Questions with rated responses	23
Rated chatbot responses	276
Readability measures	Flesch-Kincaid Grade Level and Dale-Chall familiar-vocabulary score.
Empathy measure	Average of OpenAI and Gemini LLM-judge empathy scores.
Main FKGL model	average_empathy ~ fk_grade + fk_grade^2 + llm + prompt_id + question_id fixed effects.
Main Dale-Chall model	average_empathy ~ dale_chall + llm + prompt_id + question_id fixed effects.
Design-validation model	Readability score ~ llm * prompt_id + random intercept for question.

FKGL fixed-effects model equation

$$\text{Empathy}_{iq} = \beta_0 + \beta_1 \text{FKGL}_{iq} + \beta_2 \text{FKGL}_{iq}^2 + \gamma_{m(i)} + \delta_{p(i)} + \alpha_q + \varepsilon_{iq}$$

$\gamma_{m(i)}$: LLM fixed effects (model used to generate the response)

$\delta_{p(i)}$: prompt template fixed effects

α_q : question fixed effects (conversation/question context)

ε_{iq} : error term

Dale-Chall fixed-effects model equation

$$\text{Empathy}_{iq} = \beta_0 + \beta_1 \text{DaleChall}_{iq} + \gamma_{m(i)} + \delta_{p(i)} + \alpha_q + \varepsilon_{iq}$$

$\gamma_{m(i)}$: LLM fixed effects
 $\delta_{p(i)}$: prompt template fixed effects
 α_q : question fixed effects
 ε_{iq} : error term

Variable definitions

Empathy (*average_empathy*): LLM-persona rating of response empathy on a **1–5 Likert scale**, using a rubric (higher = more empathetic).

FKGL (*fk_grade_quanteda*): Flesch–Kincaid Grade Level computed from the response text (higher = more complex reading level).

FKGL2: Quadratic term allowing a nonlinear relationship between FKGL and empathy.

DaleChall (*dale_chall_quanteda*): Dale–Chall readability score computed from the response text (higher = greater lexical difficulty / more unfamiliar words).

LLM fixed effects (*factor(llm)*): Controls for systematic differences in empathy attributable to the underlying model (e.g., Claude vs Gemini vs GPT).

Prompt fixed effects (*factor(prompt_id)*): Controls for systematic differences due to prompt template (style constraints, tone instructions, length limits).

Question fixed effects (*factor(conversation_id)*): Controls for baseline empathy differences across the 25 different frequently asked customer questions (context/need) provided by ODI.

Predicting Empathy (HC3 Robust SEs)

Question fixed effects included but not shown. Robust (HC3) standard errors reported.

Coefficient Estimates

Model	Term	Estimate	Std. Error	t value	p-value
FKGL FE (HC3)	(Intercept)	3.39	0.47	7.28	0.00
FKGL FE (HC3)	fk_grade	0.25	0.08	3.13	0.00

FKGL FE (HC3)	I(fk_grade^2)	-0.01	0.00	-2.75	0.01
FKGL FE (HC3)	factor(llm)google/gemini-2.5-fl ash	-0.60	0.10	-5.69	0.00
FKGL FE (HC3)	factor(llm)openai/gpt-4.1-mini	-0.07	0.07	-0.92	0.36
FKGL FE (HC3)	factor(prompt_id)1	-0.04	0.12	-0.32	0.75
FKGL FE (HC3)	factor(prompt_id)2	0.04	0.09	0.43	0.66
FKGL FE (HC3)	factor(prompt_id)3	0.28	0.20	1.45	0.15
Dale-Chall FE (HC3)	(Intercept)	5.02	0.22	22.51	0.00
Dale-Chall FE (HC3)	dale_chall	-0.01	0.01	-1.35	0.18
Dale-Chall FE (HC3)	factor(llm)google/gemini-2.5-fl ash	-0.72	0.09	-7.90	0.00
Dale-Chall FE (HC3)	factor(llm)openai/gpt-4.1-mini	-0.03	0.08	-0.44	0.66
Dale-Chall FE (HC3)	factor(prompt_id)1	-0.12	0.11	-1.14	0.26
Dale-Chall FE (HC3)	factor(prompt_id)2	0.04	0.10	0.37	0.71
Dale-Chall FE (HC3)	factor(prompt_id)3	-0.02	0.15	-0.16	0.87

Model Fit

Model	N	R_squared	Adj_R_squared	Residual_SE
FKGL FE (HC3)	276	0.80	0.78	0.51

Dale-Chall FE (HC3)	276	0.79	0.76	0.52
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Appendix B.6.2 Sensitivity Analysis: Drop Influential Observations (HC3 Robust SEs)

Question fixed effects included but not shown. Rows dropped are the four highest Cook's-distance rows for each model.

Coefficient Estimates

Model	Term	Estimate	Std. Error	t value	p-value
FKGL FE (Drop Influential, HC3)	(Intercept)	3.50	0.41	8.48	0.00
FKGL FE (Drop Influential, HC3)	fk_grade	0.23	0.07	3.43	0.00
FKGL FE (Drop Influential, HC3)	I(fk_grade^2)	-0.01	0.00	-3.11	0.00
FKGL FE (Drop Influential, HC3)	factor(llm)google/gemini-2.5-flash	-0.63	0.10	-6.32	0.00
FKGL FE (Drop Influential, HC3)	factor(llm)openai/gpt-4.1-mini	-0.10	0.07	-1.39	0.17
FKGL FE (Drop Influential, HC3)	factor(prompt_id)1	-0.06	0.11	-0.55	0.58
FKGL FE (Drop Influential, HC3)	factor(prompt_id)2	0.02	0.08	0.29	0.77
FKGL FE (Drop Influential, HC3)	factor(prompt_id)3	0.23	0.18	1.28	0.20
Dale-Chall FE (Drop Influential, HC3)	(Intercept)	4.94	0.20	25.20	0.00

Dale-Chall FE (Drop Influential, HC3)	dale_chall	-0.01	0.01	-0.90	0.37
Dale-Chall FE (Drop Influential, HC3)	factor(llm)google/gemini-2.5-flash	-0.74	0.08	-8.77	0.00
Dale-Chall FE (Drop Influential, HC3)	factor(llm)openai/gpt-4.1-mini	-0.07	0.07	-0.94	0.35
Dale-Chall FE (Drop Influential, HC3)	factor(prompt_id)1	-0.14	0.09	-1.50	0.13
Dale-Chall FE (Drop Influential, HC3)	factor(prompt_id)2	0.02	0.08	0.19	0.85
Dale-Chall FE (Drop Influential, HC3)	factor(prompt_id)3	-0.11	0.13	-0.83	0.41

Model Fit

Model	N	R_squared	Adj_R_squared	Residual_SE
FKGL FE (Drop Influential, HC3)	272	0.82	0.80	0.48
Dale-Chall FE (Drop Influential, HC3)	272	0.82	0.79	0.48

Appendix B.6.3 FKGL Model with Cluster-Robust Standard Errors

Question fixed effects included but not shown. Standard errors clustered by question_id.

Coefficient Estimates

Model	Term	Estimate	Std. Error	t value	p-value
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FKGL FE (Clustered by Question)	(Intercept)	3.39	0.57	5.99	0.00
FKGL FE (Clustered by Question)	fk_grade	0.25	0.10	2.60	0.01
FKGL FE (Clustered by Question)	I(fk_grade^2)	-0.01	0.00	-2.39	0.02
FKGL FE (Clustered by Question)	factor(llm)google/gemini-2.5- flash	-0.60	0.15	-4.01	0.00
FKGL FE (Clustered by Question)	factor(llm)openai/gpt-4.1-mini	-0.07	0.09	-0.77	0.44
FKGL FE (Clustered by Question)	factor(prompt_id)1	-0.04	0.10	-0.38	0.70
FKGL FE (Clustered by Question)	factor(prompt_id)2	0.04	0.06	0.69	0.49
FKGL FE (Clustered by Question)	factor(prompt_id)3	0.28	0.21	1.34	0.18

Model Fit

Model	N	R_squared	Adj_R_squared	Residual_SE
FKGL FE (Clustered by Question)	276	0.80	0.78	0.51

Appendix B.6.4 Dale-Chall Model with Cluster-Robust Standard Errors

Question fixed effects included but not shown. Standard errors clustered by question_id.

Coefficient Estimates

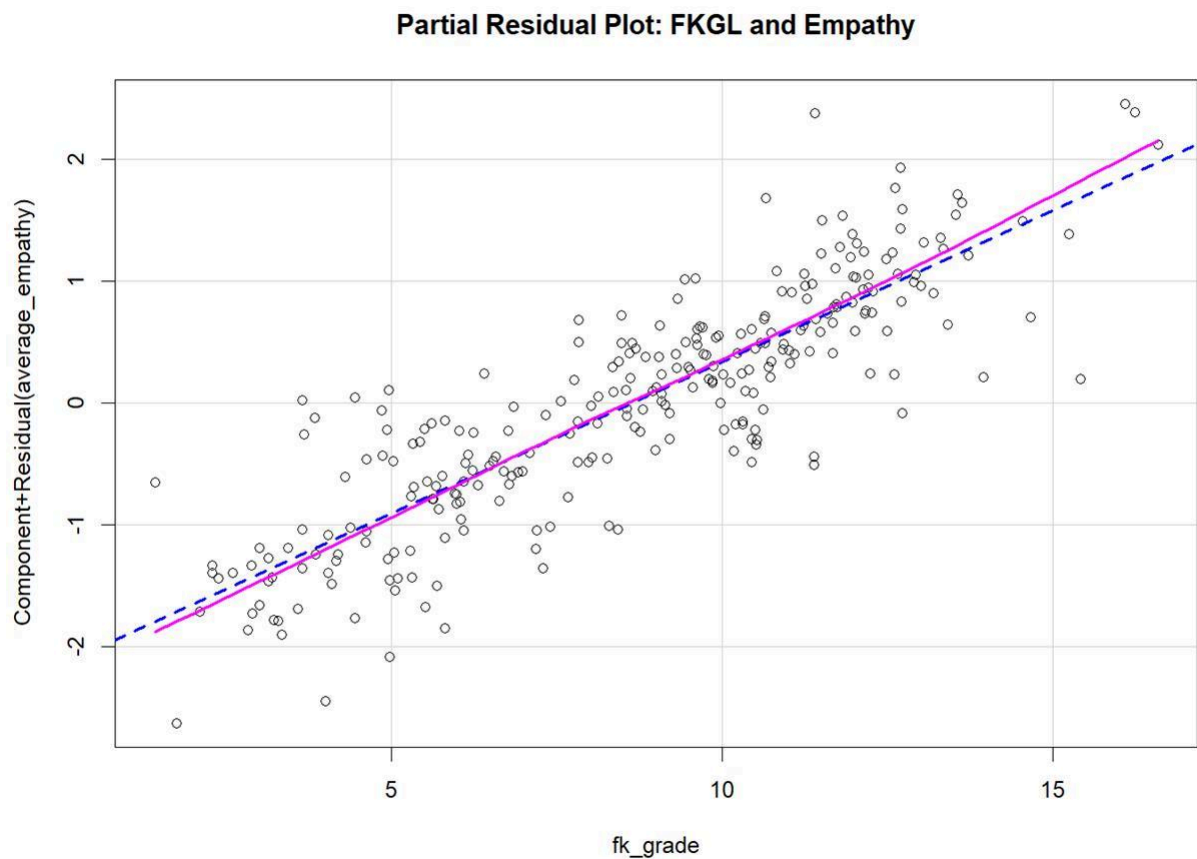
Model	Term	Estimate	Std. Error	t value	p-value
Dale-Chall FE (Clustered by Question)	(Intercept)	5.02	0.21	23.51	0.00
Dale-Chall FE (Clustered by Question)	dale_chall	-0.01	0.01	-23	0.22
Dale-Chall FE (Clustered by Question)	factor(llm)google/gemini-2.5 -flash	-0.72	0.12	-5.82	0.00
Dale-Chall FE (Clustered by Question)	factor(llm)openai/gpt-4.1-mi ni	-0.03	0.09	-0.38	0.71
Dale-Chall FE (Clustered by Question)	factor(prompt_id)1	-0.12	0.09	-1.44	0.15
Dale-Chall FE (Clustered by Question)	factor(prompt_id)2	0.04	0.05	0.66	0.51
Dale-Chall FE (Clustered by Question)	factor(prompt_id)3	-0.02	0.14	-0.16	0.87

Model Fit

Model	N	R_squared	Adj_R_squared	Residual_SE
Dale-Chall FE (Clustered by Question)	276	0.79	0.76	0.52

Appendix B.6.5 FKGL Partial Residual Plot

Component-plus-residual plot after controlling for LLM, prompt, and question fixed effects.



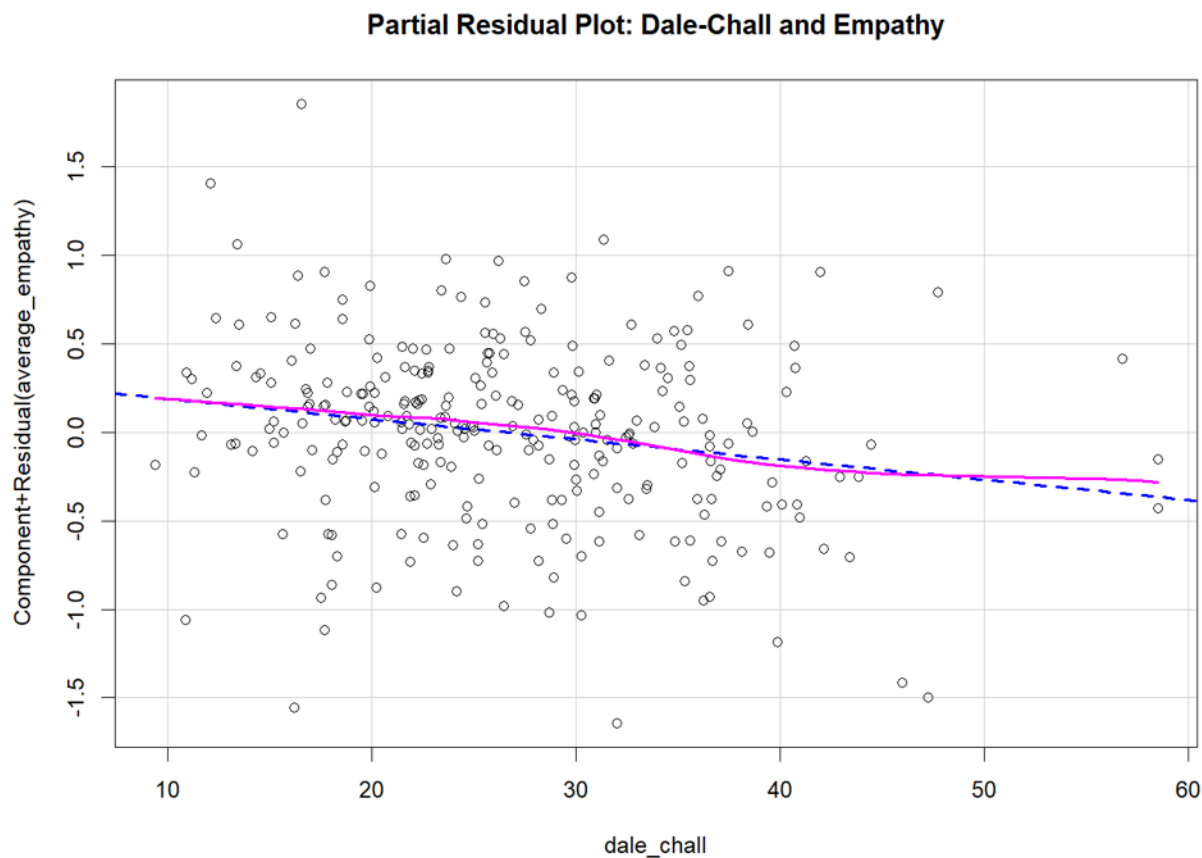
Appendix B.6.6 FKGL Linear Model vs Natural Spline

Tests whether a flexible FKGL specification improves fit relative to a linear FKGL model.

Model	Comparison	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
FKGL linear vs natural spline	1	247.00	66.68				
FKGL linear vs natural spline	2	245.00	63.12	2.00	3.57	6.92	0.00

Appendix B.6.7 Dale-Chall Partial Residual Plot

Component-plus-residual plot after controlling for LLM, prompt, and question fixed effects.



Appendix B.6.8 Dale-Chall Linear Model vs Natural Spline

Tests whether a flexible Dale-Chall specification improves fit relative to a linear Dale-Chall model.

Model	Comparison	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
Dale-Chall linear vs natural spline	1	247.00	67.17				
Dale-Chall linear vs natural spline	2	245.00	67.09	2.00	0.08	0.15	0.86

Appendix B.6.9 Collinearity Diagnostics (VIF / GVIF)

For factor variables, use the adjusted GVIF column when present.

Model	Term	GVIF	Df	GVIF ^{1/(2*Df)}
FKGL (centered quadratic)	fk_c	4.60	1.00	2.14
FKGL (centered quadratic)	I(fk_c^2)	1.30	1.00	1.14
FKGL (centered quadratic)	factor(llm)	1.59	2.00	1.12
FKGL (centered quadratic)	factor(prompt_id)	4.27	3.00	1.27
FKGL (centered quadratic)	factor(question_id)	1.41	22.00	1.01
Dale-Chall (linear)	dale_chall	2.68	1.00	1.64
Dale-Chall (linear)	factor(llm)	1.08	2.00	1.02
Dale-Chall (linear)	factor(prompt_id)	2.04	3.00	1.13
Dale-Chall (linear)	factor(question_id)	1.56	22.00	1.01

Appendix B.6.10 Heteroskedasticity Diagnostics (Breusch-Pagan Test)

Significant results motivate HC3 robust standard errors in the main tables.

Model	BP Statistic	Degrees of Freedom	p-value
FKGL FE model	61.87	29.00	0.00
Dale-Chall FE model	54.81	28.00	0.00

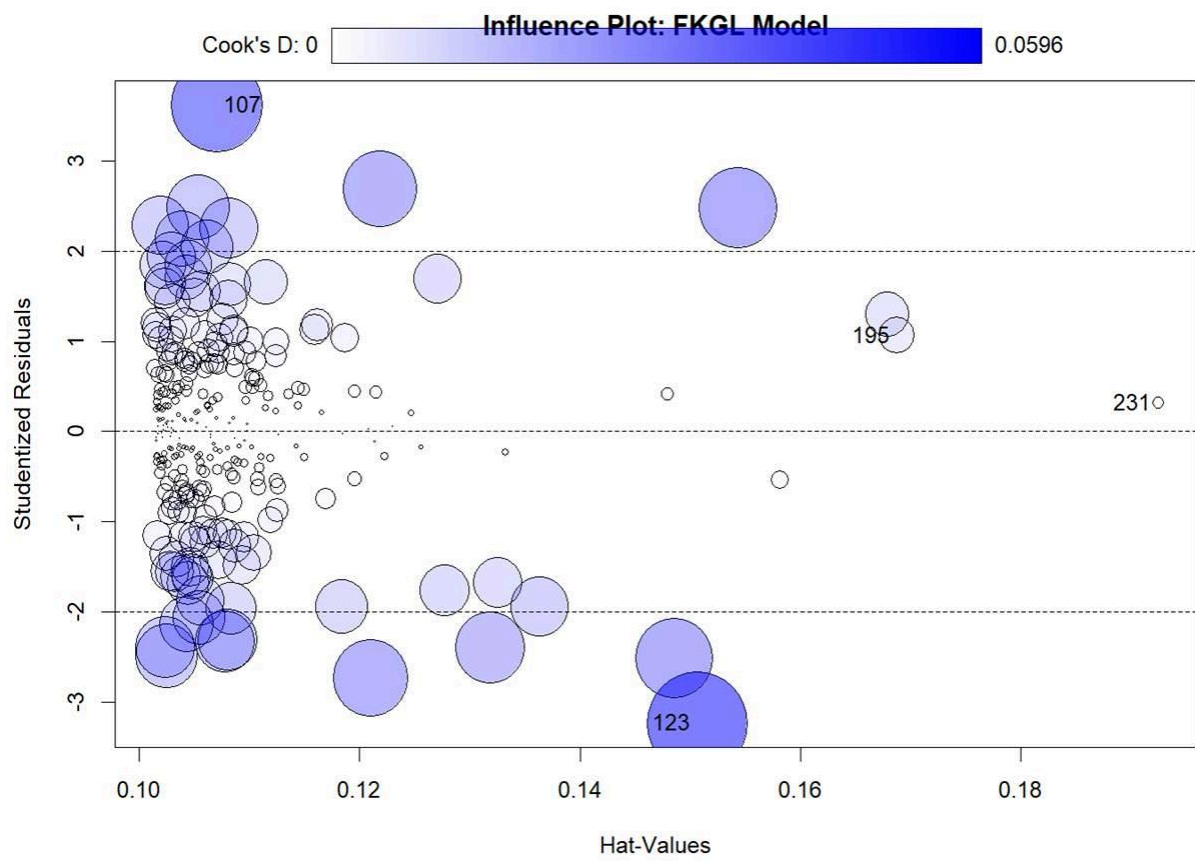
Appendix B.6.11 Influence Diagnostics

Rows are ranked by Cook's distance. Sensitivity models drop the top four rows for each model.

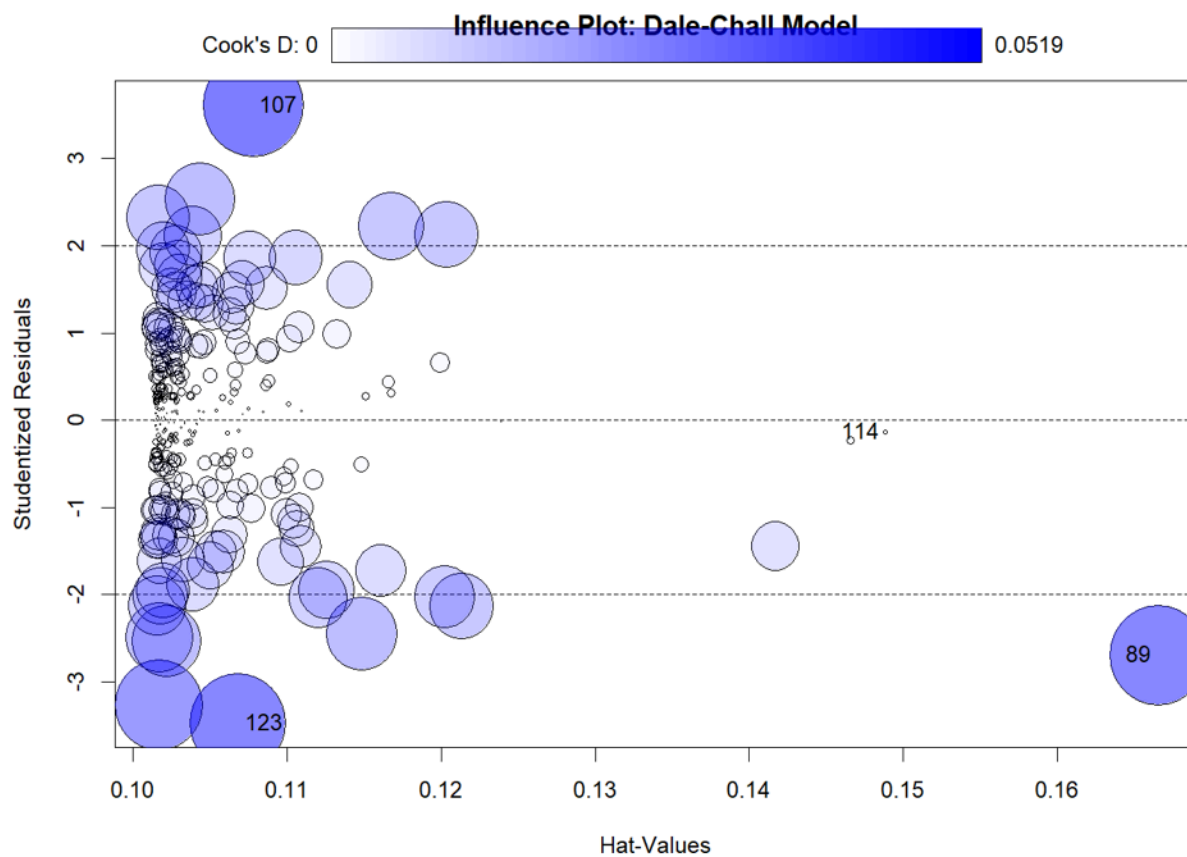
Model	Row	Cooks_D	Hat	Studentized_Residual
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FKGL FE model	123	0.06	0.15	-3.23
FKGL FE model	107	0.05	0.11	3.62
FKGL FE model	248	0.04	0.15	2.48
FKGL FE model	89	0.04	0.15	-2.51
FKGL FE model	84	0.03	0.12	-2.74
FKGL FE model	126	0.03	0.12	2.69
FKGL FE model	110	0.03	0.13	-2.40
FKGL FE model	257	0.02	0.11	2.49
FKGL FE model	115	0.02	0.10	-2.50
FKGL FE model	48	0.02	0.10	-2.39
Dale-Chall FE model	107	0.05	0.11	3.61
Dale-Chall FE model	89	0.05	0.17	-2.69
Dale-Chall FE model	123	0.05	0.11	-3.48
Dale-Chall FE model	84	0.04	0.10	-3.26
Dale-Chall FE model	186	0.03	0.11	-2.46
Dale-Chall FE model	257	0.03	0.10	2.54
Dale-Chall FE model	255	0.02	0.10	-2.54
Dale-Chall FE model	115	0.02	0.10	-2.49
Dale-Chall FE model	126	0.02	0.12	2.22
Dale-Chall FE model	48	0.02	0.12	-2.14

FKGL influence plot



Dale-Chall influence plot

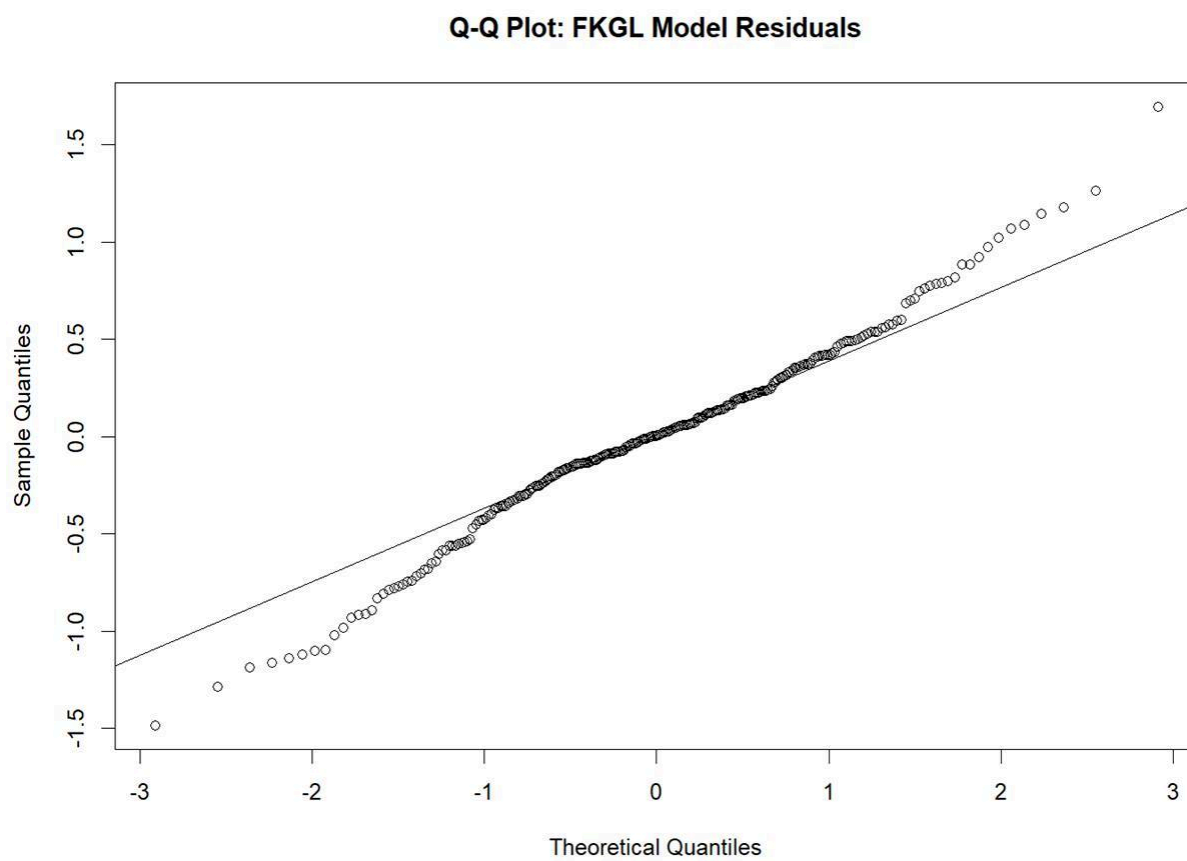


Appendix B.6.12 Residual Normality Diagnostics

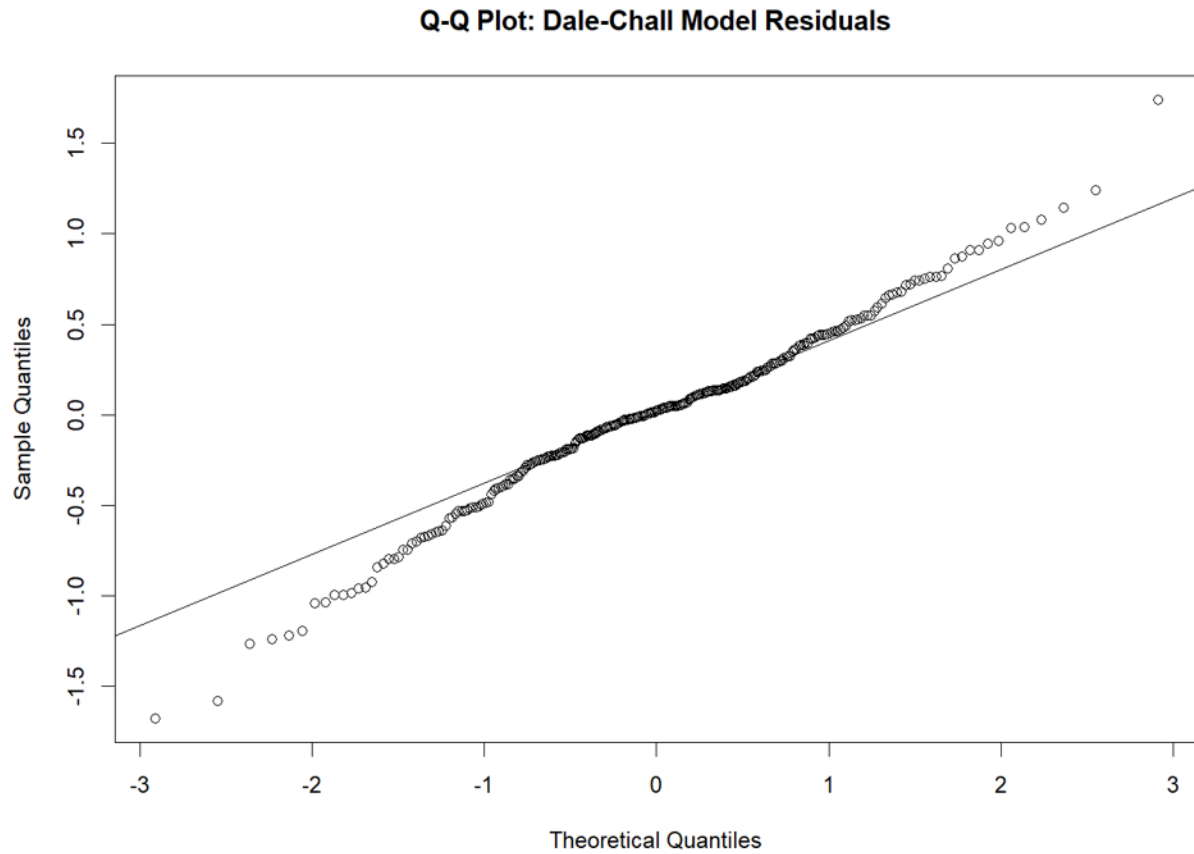
Shapiro-Wilk tests and Q-Q plots are reported. Robust standard errors are used for inference.

Model	W Statistic	p-value
FKGL FE model	0.99	0.01
Dale-Chall FE model	0.99	0.01

FKGL model Q-Q plot



Dale-Chall model Q-Q plot



Appendix B.6.13 Subgroup Results by LLM: FKGL Model

HC3 robust standard errors reported within each LLM subgroup.

Model	Term	Estimate	Std. Error	t value	p-value
anthropic/claude-sonnet-4.5	fk_grade	-0.05	0.14	-0.38	0.71
anthropic/claude-sonnet-4.5	I(fk_grade^2)	0.00	0.01	0.05	0.96
google/gemini-2.5-flash	fk_grade	0.26	0.19	1.38	0.17
google/gemini-2.5-flash	I(fk_grade^2)	-0.01	0.01	-0.96	0.34
openai/gpt-4.1-mini	fk_grade	-0.12	0.22	-0.55	0.59
openai/gpt-4.1-mini	I(fk_grade^2)	0.01	0.01	0.59	0.56

Appendix B.6.14 Subgroup Results by LLM: Dale-Chall Model

HC3 robust standard errors reported within each LLM subgroup.

Model	Term	Estimate	Std. Error	t value	p-value
anthropic/claude-sonnet-4.5	dale_chall	0.01	0.02	0.62	0.54
google/gemini-2.5-flash	dale_chall	-0.02	0.02	-0.96	0.34
openai/gpt-4.1-mini	dale_chall	0.01	0.03	0.38	0.71

Appendix B.6.14 Readability Manipulation Mixed-Model Coefficients

Models include LLM, prompt, their interaction, and a random intercept for question.

Model	Term	Estimate	Std. Error	df	t value	Pr(> t)
FKGL mixed model	(Intercept)	11.41	0.35	192.71	32.74	0.00
FKGL mixed model	llmgoogle/gemini-2.5-flash	-1.46	0.44	253.00	-3.31	0.00
FKGL mixed model	llmopenai/gpt-4.1-mini	-0.59	0.44	253.00	-1.35	0.18
FKGL mixed model	prompt_id1	-1.65	0.44	253.00	-3.75	0.00
FKGL mixed model	prompt_id2	1.29	0.44	253.00	2.93	0.00
FKGL mixed model	prompt_id3	-6.15	0.44	253.00	-13.92	0.00

FKGL mixed model	llmgoogle/gemini-2.5-flas h:prompt_id1	-2.84	0.62	253.00	-4.55	0.00
FKGL mixed model	llmopenai/gpt-4.1-mini:pr ompt_id1	-0.13	0.62	253.00	-0.22	0.83
FKGL mixed model	llmgoogle/gemini-2.5-flas h:prompt_id2	-1.81	0.62	253.00	-2.90	0.00
FKGL mixed model	llmopenai/gpt-4.1-mini:pr ompt_id2	-1.16	0.62	253.00	-1.85	0.07
FKGL mixed model	llmgoogle/gemini-2.5-flas h:prompt_id3	0.14	0.62	253.00	0.22	0.82
FKGL mixed model	llmopenai/gpt-4.1-mini:pr ompt_id3	0.44	0.62	253.00	0.70	0.48
Dale-Chall mixed model	(Intercept)	20.84	1.34	127.23	15.59	0.00
Dale-Chall mixed model	llmgoogle/gemini-2.5-flas h	2.25	1.55	253.00	1.45	0.15
Dale-Chall mixed model	llmopenai/gpt-4.1-mini	2.09	1.55	253.00	1.35	0.18
Dale-Chall mixed model	prompt_id1	2.29	1.55	253.00	1.48	0.14
Dale-Chall mixed model	prompt_id2	-1.89	1.55	253.00	-1.22	0.22
Dale-Chall mixed model	prompt_id3	14.66	1.55	253.00	9.45	0.00
Dale-Chall mixed model	llmgoogle/gemini-2.5-flas h:prompt_id1	6.28	2.19	253.00	2.86	0.00
Dale-Chall mixed model	llmopenai/gpt-4.1-mini:pr ompt_id1	1.25	2.19	253.00	0.57	0.57

Dale-Chall mixed model	llmgoogle/gemini-2.5-flan-hypothetical:pr h:prompt_id2	2.99	2.19	253.00	1.36	0.17
Dale-Chall mixed model	llmopenai/gpt-4.1-mini:pr ompt_id2	2.06	2.19	253.00	0.94	0.35
Dale-Chall mixed model	llmgoogle/gemini-2.5-flan-hypothetical:pr h:prompt_id3	-4.07	2.19	253.00	-1.85	0.07
Dale-Chall mixed model	llmopenai/gpt-4.1-mini:pr ompt_id3	-0.49	2.19	253.00	-0.22	0.82

Appendix B.6.15 Readability Manipulation Mixed-Model ANOVA

Tests whether LLM, prompt, and their interaction systematically shift readability.

Model	Comparison	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
FKGL mixed model	llm	323.24	161.62	2	253.00	72.14	0.00
FKGL mixed model	prompt_id	1,748.74	582.91	3	253.00	260.18	0.00
FKGL mixed model	llm:prompt_id	92.94	15.49	6	253.00	6.91	0.00
Dale-Chall mixed model	llm	643.94	321.97	2	253.00	11.62	0.00
Dale-Chall mixed model	prompt_id	8,084.35	2,694.78	3	253.00	97.29	0.00

Dale-Chall mixed model	llm:prompt_id	765.34	127.56	6	253.00	4.60	0.00
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Appendix B.6.16 LLM-Judge Agreement: Quadratically Weighted Kappa

Weighted kappa is calculated separately for the four SERVQUAL-inspired empathy items.

Item	N	Weighted_Kappa
individual_attention	276	0.74
caring_tone	276	0.66
customer_first_orientation	276	0.77
need_understanding	276	0.79

Appendix B.6.17

Prompt 0 (P0) - Baseline (No Style Instruction Given)

<TASK>

You are an ODI grip specialist. Recommend the most suitable ODI grip based strictly on the uploaded CSV dataset and the retrieved RAG context.

<DATA ACCESS (STRICT)>

- Use ONLY information from the retrieved RAG context.
- Do NOT use outside knowledge.
- Do NOT invent specifications or claims.

<STYLE>

<DECISION RULE>

- You MUST recommend EXACTLY ONE product for every question.
- The recommended product_name MUST appear in the current RAG context.
- If the match is weak, still pick the closest match from the RAG context.
- If question contains information on diameter (hand size, size of the grip etc), riding style (downhill, casual etc) or riding conditions (weather, season etc), check RAG context and make recommendation. If none are present, ask clarifying question.

<OUTPUT RULES>

Prompt 1 (P1) - Short Sentences

<TASK>

You are an ODI grip specialist. Recommend the most suitable ODI grip based strictly on the uploaded CSV dataset and the retrieved RAG context.

<DATA ACCESS (STRICT)>

- Use ONLY information from the retrieved RAG context.
- Do NOT use outside knowledge.
- Do NOT invent specifications or claims.

<STYLE>

Provide reasoning in short sentences.

<DECISION RULE>

- You MUST recommend EXACTLY ONE product for every question.
- The recommended product_name MUST appear in the current RAG context.
- If the match is weak, still pick the closest match from the RAG context.

- If question contains information on diameter (hand size, size of the grip etc), riding style (downhill, casual etc) or riding conditions (weather, season etc), check RAG context and make recommendation. If none are present, ask clarifying question.

<OUTPUT RULES>

Prompt 2 (P2) - Short Paragraph

<TASK>

You are an ODI grip specialist. Recommend the most suitable ODI grip based strictly on the uploaded CSV dataset and the retrieved RAG context.

<DATA ACCESS (STRICT)>

- Use ONLY information from the retrieved RAG context.
- Do NOT use outside knowledge.
- Do NOT invent specifications or claims.

<STYLE>

Provide reasoning in short paragraphs. Use maximum 4 sentences.

<DECISION RULE>

- You MUST recommend EXACTLY ONE product for every question.
- The recommended product_name MUST appear in the current RAG context.
- If the match is weak, still pick the closest match from the RAG context.
- Do NOT ask clarifying questions.

<OUTPUT RULES>

Prompt 3 (P3) - 8th Grade level

<TASK>

You are an ODI grip specialist. Recommend the most suitable ODI grip based strictly on the uploaded CSV dataset and the retrieved RAG context.

<DATA ACCESS (STRICT)>

- Use ONLY information from the retrieved RAG context.
- Do NOT use outside knowledge.
- Do NOT invent specifications or claims.

<STYLE>

Write at a 8th-grade reading level.

- Use short sentences (no more than 15 words per sentence).
- Use common, everyday words.
- Avoid technical terms unless absolutely necessary.
- Use simple sentence structure (subject–verb–object).

<DECISION RULE>

- You MUST recommend EXACTLY ONE product for every question.
- The recommended product_name MUST appear in the current RAG context.
- If the match is weak, still pick the closest match from the RAG context.
- Do NOT ask clarifying questions.

<OUTPUT RULES>

B.6.18 Prompt for Customer Ratings Studio “Step 2: Define the rubric” ([app link](#))

Role

You are a service quality evaluation expert. You are assessing sales support provided by the supporter to the customer.

Task

Refer to the context and rate content based on the items specified in Construct Definition. Rate each item on a 5-point Likert scale.

Construct Definition

Empathy is the degree to which the response makes the customer feel understood, cared for, and personally helped through its wording and tailoring.

Empathy is composed of the below items:

- individual_attention: The response refers to the customer’s specific situation, constraints, or use case rather than giving a generic answer.
- caring_tone: The response uses a supportive, reassuring, non-dismissive tone and acknowledges any discomfort, worry, or frustration expressed by the customer.
- customer_first_orientation: The response prioritizes the customer’s comfort, safety, fit, or stated goals rather than sounding salesy, scripted, or product-first.
- need_understanding: The response accurately identifies the customer’s stated need or problem and explains how the recommendation addresses it.

Inclusion Criteria

individualized_attention includes:

- Explicit references to customer's specific use case, preferences or needs
- caring_tone includes:
- Explicit use of reassuring vocabulary when the context contains emotional load such as pain, discomfort, frustration
- customer_first_orientation includes:
- Appearance of product details and customer's requirements together
- need_understanding
- Explicit acknowledgement of the need, preference or problem

Exclusion Criteria

Do not penalize individualized_attention if:

- The context does not have any details on the customer's specific use case, preferences or needs

Do not penalize caring_tone if:

- The context does not carry emotional load

Do not penalize customer_first_orientation if:

- The context does not have any details on the customer's specific use case, preferences or needs

Do not penalize need_understanding if:

- The context does not have any details on the customer's specific use case, preferences or needs

Constraints

- average_empathy_score: mean of the 4 dimension scores
- Item Scores: integers 1-5
- Item Reasoning: 1-2 short sentences per item
- Output: valid JSON only, no markdown or extra text

- Output Format

```
{
  "individual_attention_score": 1,
  "individual_attention_reasoning": "Reason here.",
  "caring_tone_score": 1,
  "caring_tone_reasoning": "Reason here.",
  "customer_first_orientation_score": 1,
  "customer_first_orientation_reasoning": "Reason here.",
  "need_understanding_score": 1,
  "need_understanding_reasoning": "Reason here.",
  "average_empathy_score": 1.0
}
```

Examples

```
{
  "examples": [
```

```

{
  "item": "individual_attention",
  "definition": "The brand employees give customers individual attention",
  "example": {
    "context": "I'll be riding in Whistler and the forecast says it might be rainy. What do you recommend?",
    "content": "This grip has an aggressive finger traction pattern and ribbed palm that adds extra surface for you to hold on better."
  },
  "annotation": {
    "what_happened": "The supporter fulfills the customer's informational need, but it is overly focused on listing the product specs instead of tying them back to the customer's intent. It appears like a generic sales pitch and lacks a personalized feel.",
    "more_empathic_alternative": "This grip has an aggressive finger traction pattern designed to make you hold on better when wet. The ribbed palm adds extra surface for stronger grip. These features will help safer riding under rainy conditions in Whistler.",
    "why_more_empathic": "This approach highlights product features while acknowledging the customer's specific use case. This makes the customer feel they are getting personalized recommendations."
  }
},
{
  "item": "caring_fashion",
  "definition": "The brand employees deal with customers in a caring fashion",
  "example": {
    "context": "I'm looking for grips that are durable. I don't want to spend too much money.",
    "content": "This is a durable grip. It is also at an affordable price point at $12.99."
  },
  "annotation": {
    "what_happened": "The supporter fulfills the customer's informational need, but is too direct and brief. This makes it appear uncaring and dismissive.",
    "more_empathic_alternative": "Sure, I can help you find durable grips that are affordable at the same time.",
    "why_more_empathic": "This approach fulfills the customer's informational need while also ensuring the customer knows they have the attention of the support rep."
  }
},
{
  "item": "best_interest_at_heart",
  "definition": "The brand employees have the customer best interest at heart",
  "example": {
    "context": "Large grips are too bulky and they make me frustrated. But I have large hands, what can I do?",
    "content": "This is a slim grip that suits large hands and offers padding without being bulky."
  }
}

```



```

    },
    "annotation": {
      "what_happened": "The supporter fulfills the customer's informational need and addresses their individual requirements, but it does not establish a personal connection. The customer asks an emotionally loaded question, expressing frustration, but the support does not acknowledge the emotion. It makes it look like support is prioritizing a sale rather than the customer's comfort.",
      "more_empathic_alternative": "I understand bulkiness can be frustrating. Here is a slimmer alternative that offers high damping, and balances comfort and traction. This hybrid design helps keep riding enjoyable and comfortable for larger hands.",
      "why_more_empathic": "This approach highlights product features while acknowledging the customer's emotional state and preferences. This makes the customer feel reassured that the support has their best interest at heart."
    }
  },
  {
    "item": "understand_customer_needs",
    "definition": "The brand employees understand the needs of their customers",
    "example": {
      "context": "My hands hurt after a while with my current grips.",
      "content": "This grip has thick padding."
    },
    "annotation": {
      "what_happened": "The supporter makes a recommendation with a relevant feature, but it does not acknowledge the customer's specific need. This shows a lack of insight and understanding, looking like a generic RAG response.",
      "more_empathic_alternative": "I understand you may need extra padding. This grip has a high damping level, which means thicker padding to reduce hand pain. This grip addresses your hand pain concern with a medium diameter and hybrid offset design that provides extra padding on the palm side, where you're experiencing discomfort between the thumb and pointer.",
      "why_more_empathic": "This approach highlights product features while acknowledging the customer's specific need and how the product feature relates to the need. This indicates insight and initiative."
    }
  }
]
}

```

B.6.19 Prompt for Customer Ratings Studio “Step 4: Review disagreements” ([app link](#))

#Construct Definition (Context)

Empathy is the degree to which the response makes the customer feel understood, cared for, and personally helped through its wording and tailoring.

Empathy is composed of the below items:

- individual_attention: The response refers to the customer's specific situation, constraints, or use case rather than giving a generic answer.
- caring_tone: The response uses a supportive, reassuring, non-dismissive tone and acknowledges any discomfort, worry, or frustration expressed by the customer.
- customer_first_orientation: The response prioritizes the customer's comfort, safety, fit, or stated goals rather than sounding salesy, scripted, or product-first.
- need_understanding: The response accurately identifies the customer's stated need or problem and explains how the recommendation addresses it.

Role

You are a measurement-audit judge. You do not score the item yourself. You analyze disagreements between raters and recommend edits to the rating instrument to improve quadratic weighted kappa.

Task

- Understand the construct definition text.
- Read through the dataset, comparing the score and the reasoning values row by row.
- Identify the source of disagreement by referring to the reasonings provided.
- Based on your findings on the disagreements, suggest concrete refinements to the prompt, criteria, or examples that would reduce ambiguity and standardize interpretation of the construct.

Inclusion Criteria

Propose edits when you detect any of the following:

- Unclear construct definition
- Ambiguous thresholds for score levels
- Inconsistent interpretation of customer persona-specific standards
- Missing examples for important edge cases
- Conflicting rules within the instrument
- Disagreement caused by vague wording in the criteria

Exclusion Criteria

Do not propose edits that:

- Redefine the construct into a different concept
- Make the prompt longer unless the added text clearly reduces ambiguity
- Judge the row directly instead of diagnosing disagreement

Constraints

- Do not rate the item yourself
- Focus on why the raters disagreed
- Base recommendations only on the construct definition and rater outputs provided
- Suggest concrete wording changes, not vague advice

- Return valid JSON only
- Do not return markdown, code fences, or extra text

Examples

Example recommendations:

- Clarify the edge cases where a response is short but still complete
- Define what counts as too vague versus appropriately concise
- Add an example showing a borderline score and why it belongs there